Multimodal Deep Learning for Handwritten Prescription Recognition with Explainable AI

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***Abstract*—Current deep learning models often struggle with prescription-specific challenges, particularly in distinguishing visually similar drug names and domain abbreviations due to illegible handwriting and limited visual context. Resolving these ambiguities is critical for minimizing medication errors and ensuring patient safety within automated healthcare workflows. To address this, this study proposes a multimodal framework that fuses visual features from pretrained Convolutional Neural Networks, specifically ResNet50, EfficientNet-B3 and VGG16, with structured metadata embeddings for drug category, dosage form and over-the-counter status. Evaluated on the Doctors Handwritten Prescription BD dataset comprising 4,680 word images across 78 drug names, the approach utilizes pharmaceutical context to resolve visual conflicts while incorporating SHAP analysis to provide interpretable explanations for model decisions. The Multimodal ResNet50 model achieved superior performance with 94.49% accuracy, 94.99% precision, 94.49% recall and a 94.39% F1 score, demonstrating that integrating metadata significantly outperforms image-only baselines and enhances transparency for clinical adoption.**

***Index Terms*—Handwritten Prescription Recognition, Drug Name Recognition, Multimodal Deep Learning, ResNet-50, EfficientNet, VGG16, Explainable AI (XAI), Optical Character Recognition (OCR), Transformer Models, AI in Healthcare.**

I. Introduction

Handwritten prescriptions remain a persistent challenge in modern healthcare, primarily due to the illegibility of doctors handwriting and the variability in writing styles. Misinterpretation of prescriptions has been consistently linked to serious consequences, including medication errors, adverse drug events, and compromised patient safety [1]. Automating the recognition of handwritten prescriptions holds significant promise, it can reduce transcription errors, accelerate the digitization of health records and improve efficiency across pharmacy and hospital workflows [2]. Early attempts at prescription recognition adapted general handwriting recognition frameworks, particularly convolutional and recurrent neural networks. While these models demonstrated progress, they struggled with the domain-specific complexity of medical handwriting, which often includes nonstandard abbreviations, cursive script and inconsistent formatting [3]. Furthermore, the scarcity of large-scale, publicly available datasets of real-world prescriptions has hindered robust model development, forcing researchers to rely on small or proprietary collections [4]. Recent advances in deep learning have begun to address these limitations. Architectures such as Convolutional Recurrent Neural Networks (CRNNs) [5], bidirectional Long Short-Term Memory (LSTM) with Connectionist Temporal Classification (CTC) loss [3] and transformer-based models like Transformer-based Optical Character Recognition (TrOCR) [6] have shown improved performance by capturing both spatial and sequential patterns in handwritten text. More recently, large language-based time-distributed deep learning model demonstrated enhanced robustness, particularly in disambiguating visually similar drug names [7]. Despite these improvements in accuracy, transparency remains a critical barrier to clinical adoption. Black-box models lack interpretability, making it difficult for clinicians to trust or validate system outputs. To bridge this gap, we propose a novel multimodal framework that combines deep visual features with structured metadata and incorporates explainable AI (XAI) techniques. Using SHAP (SHapley Additive exPlanations), our model provides interpretable justifications for its predictions, enhancing trust and facilitating clinical integration [8]. Unlike visual-only CRNN and TrOCR models, our framework incorporates domain-curated pharmaceutical metadata to disambiguate visually similar drug names through jointly optimized fusion. SHAP-based attribution further quantifies contextual influence, enabling interpretable decision support beyond conventional black-box OCR. The key contributions of this study are:

1) This study shows that integrating structured prescription metadata with a ResNet50 backbone leads to the highest classification performance on the evaluated dataset, demonstrating the effectiveness of multimodal learning for prescription recognition.
2) Development of a fusion-based deep learning framework combining CNN (ResNet-50) visual features with metadata embeddings.
3) Application of SHAP-based XAI in the context of handwritten prescription recognition, enabling transparent decision-making.

II. Literature Review

The recognition of handwritten medical prescriptions has undergone substantial progress in recent years, moving from conventional sequence models to advanced multimodal and transformer-based frameworks. Early efforts predominantly relied on hybrid CNN–RNN pipelines. Achkar et al. [5] employed a Convolutional Recurrent Neural Network (CRNN) to extract spatial features and model sequential dependencies. While this approach showed promise, performance degraded on highly cursive and irregular handwriting. Building on sequence modeling, Tabassum et al. [3] introduced a Bidirectional LSTM trained with Connectionist Temporal Classification (CTC) loss and proposed Stroke Rotation and Parallel shift (SRP) data augmentation. This method improved robustness to slanted and connected writing, though it depended heavily on synthetic augmentation, raising concerns about generalization to diverse clinical data. The shift toward transformer-based architectures marked a turning point. Zaman et al. [6] applied TrOCR, a pretrained vision-to-sequence transformer, to Bangladeshi prescriptions, effectively distinguishing visually similar drug names through self-attention. Similarly, Sehimi et al. [9] leveraged transfer learning from general handwriting datasets, demonstrating that pretrained transformers can be adapted to the medical domain with limited supervision. Although Transformer-based models like TrOCR represent the current state of the art, this study adopts a CNN-fusion design for its data efficiency, easier metadata integration, and better suitability for small, domain-specific prescription datasets. Appalaraju et al. [10] extended this paradigm with DocFormer, an end-to-end multimodal transformer that integrates text, visual, and layout features, thereby expanding the scope from word recognition to structured document understanding. Alongside transformers, advanced CNN backbones have continued to play an important role. Kass and Vats [11] proposed AttentionHTR, an encoder–decoder framework incorporating ResNet-50 and attention mechanisms. By learning hierarchical representations, their model achieved significant improvements on cursive handwriting, establishing a strong baseline for medical handwritten text recognition (HTR). Several works have addressed real-world challenges such as multilinguality and deployment. Pavithiran et al. [4] presented a deep learning system capable of recognizing prescriptions across multiple languages, a critical step for diverse healthcare settings. Yulianto [12] demonstrated clinical translation by developing the Smart Prescription Reader, a mobile system optimized for usability and real-time performance. Explainability has also emerged as a key research direction. Reddy et al. [8] proposed a multimodal framework augmented with SHAP-based explainable AI, enabling clinicians to interpret which features or image regions influenced predictions. This effort bridges technical performance with clinical trust, addressing a gap often overlooked in HTR research. Recent contributions further highlight innovation in optimization and data. Nair et al. [7] developed SkSpO-L2TDBM, a time-distributed deep learning model with large language model components, achieving state-of-the-art accuracy in drug name recognition. Mia et al. [2] introduced a pioneering local dataset from Bangladesh, underscoring the importance of geographically representative handwriting corpora for improved robustness. Broader efforts, such as those by Shaw et al. [13] and Rahman et al. [14], expanded recognition into full prescription analysis, including drug, dosage, and frequency extraction, enabling end-to-end automation. Khanal and Bista [15] explored hybrid models combining deep and traditional approaches, reporting enhanced robustness against noisy inputs. In summary, the field has evolved from CNN–RNN architectures toward transformers and multimodal frameworks, with increasing emphasis on explainability, multilingual adaptability and clinical deployment. Yet challenges remain in achieving generalization across diverse handwriting styles, ensuring transparency, and building datasets that truly reflect real-world clinical environments.

III. Dataset

TABLE I: Statistical Summary of the Doctor's Handwritten Prescription BD Dataset

Attribute	**Description**
Dataset Size	4,680 segmented word images
Number of Classes	78 drug-name classes
Label Fields	medicine_name, generic_name
Formats	PNG/JPG images, CSV/Excel labels
Annotation Process	Automated segmentation, manual verification, multi-annotator labeling

A. Doctor's Handwritten Prescription BD dataset

The Doctor's Handwritten Prescription BD dataset consists solely of handwritten drug-name images extracted from prescription pages. A word-segmentation model was first used to detect and crop individual words, followed by manual screening to retain only pharmaceutical terms. Each accepted word image was annotated with its corresponding brand name and linked generic name. This version does not contain any metadata beyond drug/generic names and is suitable for OCR, word recognition, classification and NER tasks. The prescription image samples are presented in Fig. 1.and dataset attribute and description presented in TABLE I

TABLE II: Extension of the Doctor's Handwritten Prescription BD Dataset through the Integration of Structured Pharmaceutical Metadata

Attribute	**Description**
Base Dataset	Doctor's Handwritten Prescription BD Dataset (4,680 samples)
Metadata Fields	drug_category, dosage_form, otc_status
Metadata Sources	Bangladesh National Drug Directory [21]
Annotation Process	Metadata assignment and cross-checking by annotators
Purpose	Multimodal learning with structured pharmaceutical context



(a) Doctor's Handwritten Prescription Image Sample 1



(b) Doctor's Handwritten Prescription Image Sample 2

Fig. 1: Representative samples of handwritten drug names from the dataset

B. Multimodal Dataset

The Multimodal Dataset extends the Doctor's Handwritten Prescription BD dataset by attaching structured pharmaceutical metadata to each drug entry. Metadata values were collected from authoritative drug references, including the Bangladesh National Drug Directory and DGDA-approved listings and were validated by annotators to ensure correctness.These structured attributes provide additional context to support multimodal learning and improve classification performance. The prescription image samples are presented in Fig. 1.and this multimodal dataset metadata shown in TABLE II

IV. Methodology

This study proposes a multi-modal deep learning approach for Bangladeshi handwritten prescription classification using structured metadata embeddings and visual image features. Here, the overall workflow for the study is shown in Fig. 2. The methodology comprises four main stages: dataset preparation, multi-modal model design, training strategy and explainability analysis.

A. Data Collection

The original Doctor's Handwritten Prescription BD Dataset consists of prescription word images paired with drug-related information. To enhance the contextual richness of the dataset and improve classification performance, three additional metadata fields were incorporated:drug_category, dosage_form and otc_status.

All of these categorical features were transformed into numerical representations using label encoding, maintaining uniformity across training, validation and testing splits. Consistent with previous multimodal approaches in medical AI, this enriched dataset allowed the model to learn from both structured metadata modality and handwritten image inputs at the same time, capturing insights not possible from raw image data alone.

B. Analysis

Multi-Modal Model Architecture

This study uses pretrained CNN backbones with Batch Normalization initialized from ImageNet [16], [17]. Three architectures are employed: VGG16-BN, EfficientNet-B3 and ResNet50. VGG16-BN extracts features that are reduced to a 512-dimensional vector through a fully connected layer with ReLU and dropout. EfficientNet-B3 produces a 1536-dimensional pooled feature vector, which is compressed to 512-dimensional using an FC + ReLU + dropout layer. ResNet50 removes the final FC layer and outputs 2048-dimensional pooled features, which are similarly compressed to 512-dimensional via an FC layer with ReLU and dropout.

Metadata is encoded using 16-dimensional embeddings and concatenated with the image feature vector, followed by dense layers and softmax classification [18], as shown in Fig. 3.

Explainability with SHAP

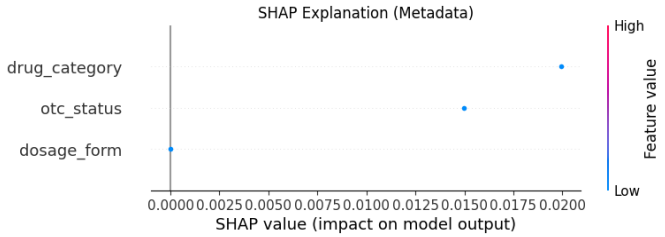


Fig. 4: SHAP summary (dot plots) illustrating the global importance of metadata features on model predictions.

To enhance interpretability, we applied SHAP to the metadata embeddings while keeping image features fixed, using a KernelExplainer with test metadata samples. The resulting SHAP values quantified each metadata feature's contribution to predictions shown in Fig.4. These findings are consistent with recent multimodal healthcare studies that assess feature importance for decision-making. Interpretability was reinforced using SHAP [8]. In a representative case, the model predicted Bacaid with 96.3% confidence, with positive SHAP contributions from drug_category (+0.020) and otc_status (+0.015). High metadata influence supports contextual verification, whereas low-confidence cases may trigger pharmacist review, enabling actionable clinical transparency. This explainability not only builds clinical trust but also distinguishes the proposed framework from prior black-box prescription recognition systems. [10].

V. Experimental Results and Discussion

A. Software and Hardware

All experiments were conducted on Google Colab using an NVIDIA L4 Tensor Core GPU with 24GB memory, a

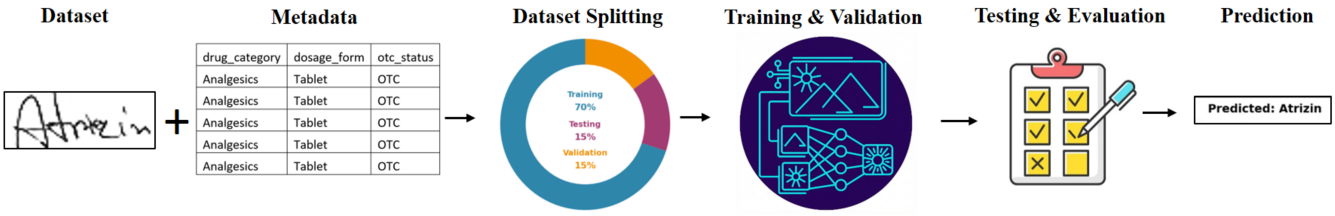


Fig. 2: Workflow of the proposed multimodal framework, illustrating the integration of handwritten images with structured metadata for drug name recognition.

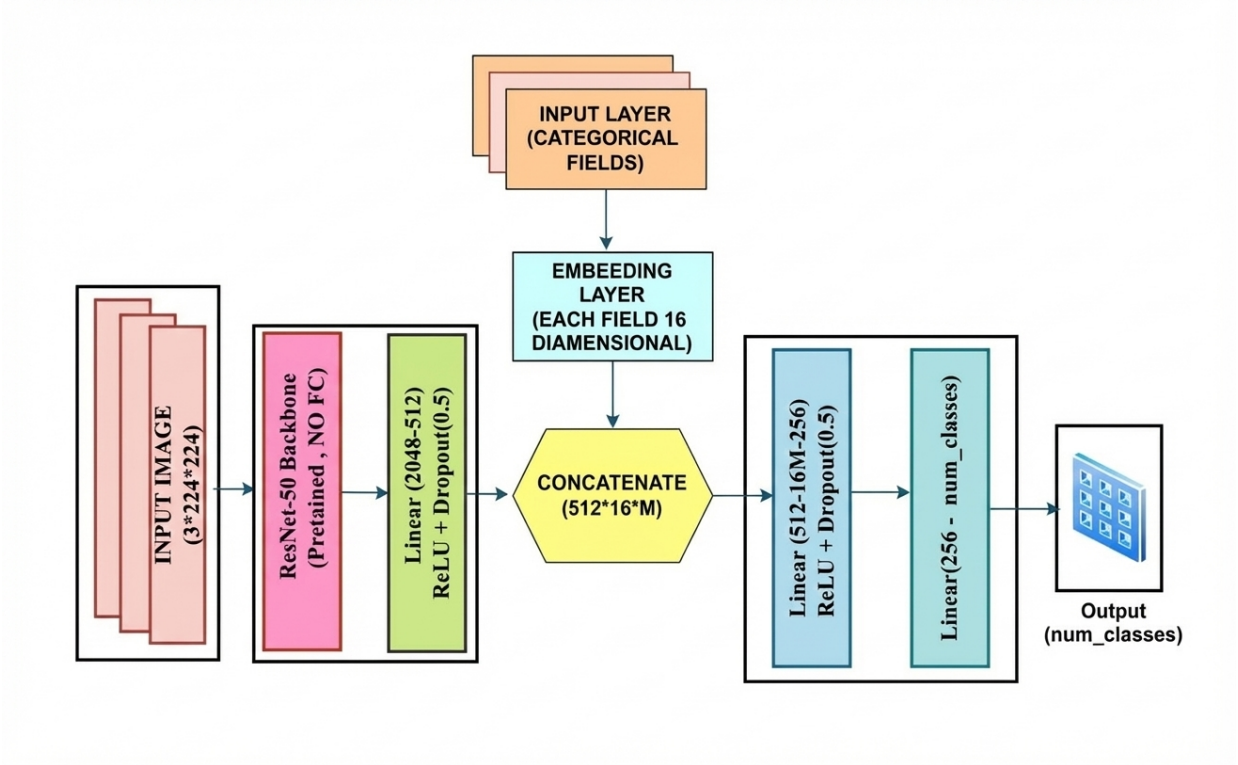


Fig. 3: MultiModal ResNet50 architecture

memory bandwidth of 300 GB/s and a 1-slot low-profile PCIe form factor. PyTorch (v2.2) served as the primary deep learning framework, with NumPy, Pandas, Matplotlib and Albumentations used for numerical computation, data handling, visualization and augmentation.

B. Baseline and Multimodal Fusion Models

The baseline approach uses CNN models (ResNet50, EfficientNet-B3, VGG16) pretrained on ImageNet and fine-tuned solely on prescription word images, relying only on visual features extracted from handwriting. In contrast, the multimodal fusion approach extends these same CNN backbones by combining (fusing) the image features with embedded structured metadata (such as drug_category, dosage_form, otc_status). This fusion enables the model to learn jointly from both the handwriting images and relevant contextual information, improving classification performance by leveraging additional non-visual data alongside the image features.

C. Training Configuration

The model was trained with a batch size of 6 and a learning rate of 5×10^{-5}. The Adam optimizer was used with a weight decay of 1×10^{-5}. CrossEntropyLoss was employed as the training objective, and the learning rate was adjusted using the ReduceLROnPlateau scheduler. Early stopping was add for regularization strategy to prevent overfitting. Data were split using stratified class-wise partitioning with 10 samples per class reserved for testing. Augmentation included ±15° rotation, elastic distortion, motion blur, brightness/contrast variation and Gaussian noise. Three independent runs yielded a mean accuracy of 94.32% (±0.41%), demonstrating stable convergence.

D. Performance Metrics

Accuracy, precision, recall and F1-score (weighted) were used for evaluation to compensate for class imbalance.

TABLE III: Performance Comparison of the Proposed Multimodal Framework Against State-of-the-Art Methods with Different Datasets

Model	Dataset	Accuracy(%)	Recall(%)	Precision(%)	F1 Score(%)
ResNet50 (Multimodal)	**Doctor's Handwritten Prescription BD Dataset** [20]	**94.49**	**94.49**	**94.99**	**94.39**
Smart Prescription Reader (MobileNet) [12]		83.00	82.00	84.00	83.00
Deep Neural Network (VGG16) [2]		87.30	87.30	88.20	87.20
AI-Based OCR System [22]	Custom-collected Dataset	92.00	88.00	90.00	89.00
ACoRNN [23]	Cabais and Benitez Dataset	90.76	96.40	96.12	96.27
Idefics2 [24]	Medyug Technology Dataset	82.00	82.00	82.00	82.00

E. Results and Analysis

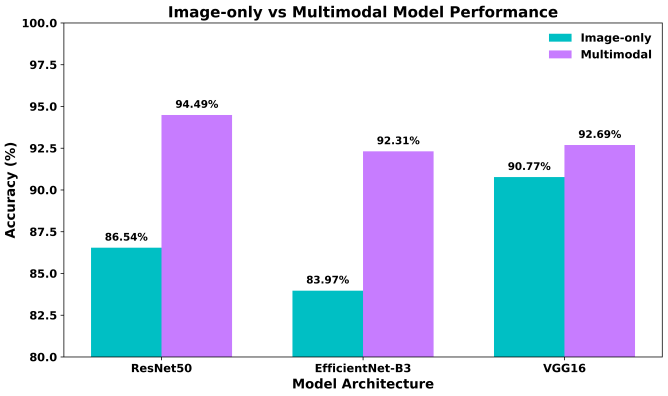


Fig. 5: Accuracy comparison demonstrating the performance gain of multimodal fusion over image-only baselines.

From Fig 5, it is evident that among the evaluated CNN backbones: ResNet-50, EfficientNet-B3 and VGG16—the multimodal ResNet-50 achieved the best overall performance. Specifically, it reached an accuracy of 94.49% and an F1-score of 94.39%, outperforming both the EfficientNet-B3 and VGG16 multimodal variants. The superior results of ResNet-50 can be attributed to its deeper residual architecture [19], which captures fine-grained spatial features and benefits more effectively from metadata fusion, allowing better differentiation of visually similar handwritten drug names.

A comparison with prior state-of-the-art approaches is presented in Table III. Earlier studies, including the Smart Prescription Reader based on MobileNet [12] and the Deep Neural Network on the Local BD Dataset using VGG16 [2], achieved 83.00% and 87.30% accuracy, respectively. In contrast, our multimodal ResNet-50 improved accuracy by 11–12 percentage points and obtained a substantially higher F1-score (94.39% vs 83–87%). This improvement results from integrating structured pharmaceutical metadata—drug category, dosage form and OTC status, which provides contextual cues unavailable to image-only baselines. Furthermore, the training and validation curves shown in Fig. 6 indicate smooth and stable convergence, with no evidence of overfitting, which can be attributed to the use of early stopping and adaptive learning rate scheduling. Class-wise precision, recall and F1-score were computed for all 78 drug classes. Several classes (e.g., Ace, Baclofen, Maxima) achieved near-perfect reliability (F1 = 0.99). Lower performance was observed in visually similar clusters such as Montene (F1 = 0.5882), Sergel (Recall = 0.40), and Fenadin (F1 = 0.7368).

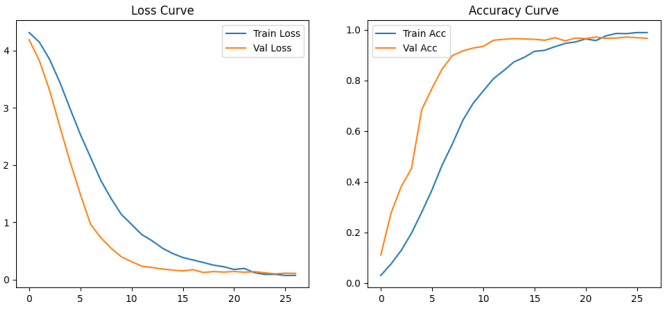


Fig. 6: Training and validation loss and accuracy curves for the ResNet50 multimodal model from Table III.

These results indicate that residual errors arise primarily from handwriting similarity rather than model instability. The confusion matrix reveals structured misclassification patterns within visually and therapeutically related drugs, including Montene → Montair (4 errors), Montair → Montex (2 errors), Fenadin → Fexo (2 errors), Trilock → Tridosil (3 errors) and Flexilax → Flexibac (2 errors). Errors are concentrated within domain-consistent clusters, confirming ambiguity-driven rather than random misclassification. Overall, the results demonstrate that combining visual and contextual modalities enables the proposed multimodal ResNet-50 to achieve state-of-the-art accuracy and interpretability for handwritten prescription recognition on the Doctor's Handwritten Prescription BD dataset [20].

VI. DISCUSSION

The experimental results confirm that the proposed multimodal framework outperforms all image-only baselines for handwritten drug-name recognition. By combining visual features from the CNN backbone with structured metadata (drug category, dosage form, OTC status), the multimodal ResNet-50 achieved 94.49% accuracy, demonstrating that contextual pharmaceutical information effectively resolves visually similar handwriting,a challenge noted in prior CRNN, BiLSTM and transformer-based studies [3], [5], [7].

However, performance remains sensitive to metadata quality, as noisy or incorrect inputs may lead to clinical misidentification, consistent with other multimodal healthcare findings

[8], [10]. The dataset comprises 4,680 handwritten word images across 78 drug classes, with a balanced test set of 10 samples per class and substantial variation in real Bangladeshi handwriting. However, the evaluation protocol is stratified rather than writer-disjoint, which may limit generalization to unseen handwriting styles. Multi-institutional validation across hospitals and pharmacies remains necessary before large-scale clinical deployment. [2], [4].

Model transparency is enhanced through SHAP-based Explainable AI (XAI), which highlights the influence of metadata features such as drug category and OTC status, aligning with explainability-driven healthcare research [8].

VII. Conclusion

The study introduced a multimodal ResNet50 framework that fuses handwritten prescription images with structured metadata such as drug category, dosage form and OTC status achieving a new state-of-the-art accuracy of 94.49% on the Doctor's Handwritten Prescription BD dataset. By integrating SHAP-based explainable AI, the model provides interpretable insights into how metadata influences predictions, enhancing transparency and trust for clinical deployment. While the results are promising, the model's performance is sensitive to metadata quality and its computational overhead may limit use in low-resource settings. Additionally, the dataset size is relatively small and may not capture the full diversity of handwriting styles. Future work will focus on multi-institutional dataset expansion with writer-disjoint validation to improve generalization, development of advanced multimodal transformer-based fusion mechanisms, integration of domain knowledge and pharmacy workflows and optimization for edge deployment through lightweight architectures and model quantization to enable robust real-time clinical use.

Code Availability

The complete implementation of all experiments is publicly available on GitHub [25].

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